Final Assignment Results (Derived from Data Analysis)

1. Market Sentiment vs. Win Rate (When do traders execute the most profitable trades?) After analyzing the data, a very interesting pattern has emerged:

- **Extreme Greed:** The highest win rate occurs here (**~89.16%**). This indicates that when the market experiences extreme greed, trades tend to be the most successful.
- **Fear:** The win rate here is **~87.28%**.
- **Neutral:** The win rate is **~82.38%**.
- **Extreme Fear & Greed (Standard):** These two sentiments showed the lowest win rates (**~76%**).

Insight: Generally, people assume that traders suffer losses during 'Extreme Greed', but according to our dataset, traders are actually closing the highest number of successful trades during this phase.

2. Profit and Loss Analysis Here is the Average Profit/Loss calculated based on each market sentiment:

- **Extreme Greed:** The average profit per trade is the highest (**$67.89**).
- **Fear:** The average profit is **$54.29**.
- **Neutral / Extreme Fear:** The average profit is the lowest here (around **$34**).

Insight: During 'Extreme Greed', not only is the probability of winning a trade higher, but the actual profit amount realized is also the largest.

3. Trading Behavior: Long (Buy) vs. Short (Sell) How does market sentiment dictate the direction of the trades?

- During days characterized by **Fear** and **Extreme Greed**, traders opened slightly more **SELL (Short)** positions compared to BUY positions.
- During **Neutral** and **Extreme Fear** days, the BUY and SELL volumes remained almost equal (balanced).

Smarter Trading Strategy

Based on these data-driven insights, I can include the following strategies at the end of my assignment:

- **Follow the Momentum (The trend is your friend):** Since both the win rate and profit margins peak during 'Extreme Greed', traders should utilize trend-following strategies during these periods. Instead of being overly cautious, traders should capitalize on the strong market momentum.

- **Avoid Choppy Markets (Neutral phase):** In the 'Neutral' or 'Extreme Fear' phases, profit margins are essentially cut in half (averaging $34). During these times, traders should practice stricter risk management by keeping their position sizing small and minimizing the use of leverage.

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